

Microeconomics In-class Quiz 2

Fall 2026

Student ID: _____ Name: _____

Instructions

1. Do NOT flip over this page until every student receives this quiz. Your TA will let you know when you can start.
2. During this closed-book quiz, you cannot consult any materials.
3. If you are unable to explain your reasoning in English, it is okay to write in Korean.
4. Should you need more space, use the backside of the page, clearly indicating the relevant quiz number.

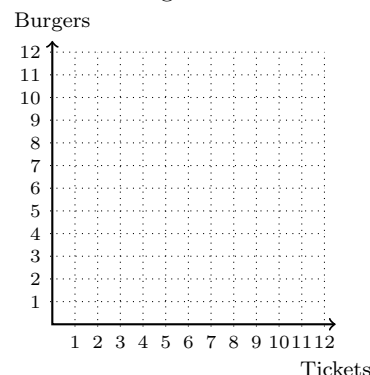
Honor Code: Cheating on exams or quizzes, plagiarizing someone else's answers as one's own, or any other instance of academic dishonesty violates the standards of academic integrity.

Confidentiality Code: Sharing the information of the exam or quiz contents with other students in any form and medium is strongly prohibited, as it raises information inequity.

I, _____, consent to the Honor Code and the Confidentiality Code.
(write your name)

1. With a weekly allowance of \$120, Sheldon buys basketball tickets and hamburgers.

- (a) Suppose a basketball ticket costs \$10, and a burger costs \$20. Draw his budget constraint on the figure to the right. Label it BC_1 .
- (b) Suppose a basketball ticket costs \$15, and a burger costs \$20. The burger place offers a discount: the price of a burger drops to \$10 *starting from the 4th one* (i.e., the first 3 burgers cost \$20 each, and any additional burgers cost \$10 each). Draw his budget constraint on the same figure. Label it BC_2 .



2. Larry's utility function is $U(X, Y) = 4X + 12Y$. The unit price for X is \$5, and the unit price for Y is \$10. Larry's income is \$50. Which of the following statements is **incorrect**?

1. For Larry, X and Y are perfect substitutes.
2. Larry has a constant marginal rate of substitution.
3. To maximize his utility, Larry will spend all his income to buy X .
4. 5 units of X and 2 units of Y is a feasible consumption bundle for Larry.

3. Chloe earns \$500 per week, spending it entirely on books and coffee. When the price of a book is \$25 and a coffee is \$5, she buys 16 books and 20 coffees. Suppose her income falls to \$450, while prices remain unchanged. She now buys 13 books and 25 coffees. Determine whether each of the following statements is True or False.

- (a) For Chloe, coffee is an inferior good. (True / False)
- (b) For Chloe, books are a luxury good. (True / False)
- (c) When her income fell to \$450, her budget constraint's slope became flatter. (True / False)

4. Suppose the production function for a small plumbing company is $Q = 2KL$, where Q is the number of water heaters installed, K is the number of machine hours, and L is the number of plumbers. The wage rate is \$20 per hour, and the rental rate of capital is \$40 per hour.

- (a) What is the marginal rate of technical substitution between labor and capital?
- (b) The company has to install 100 new heaters. What's the cost-minimizing bundle of capital and labor?

5. The following figures show several isoquants and one isocost line ($C = \$1200$) for a local taxicab company.

- (a) Infer from the isocost line ($C = \$1200$) the wage rate for a driver and the rental rate for a taxicab.
- (b) Graph the firm's expansion path on the left figure.
- (c) From the expansion path and the given costs, derive the company's long-run cost function and graph it on the right figure.

